We arrive at a value of 67 LOC for the traces instrumentation alone. Taking the two LOC numbers together, we can calculate the B2I ratio and arrive at a value of 2.79, which is almost double of logs alone (1.33) and slightly higher than metrics instrumentation alone (2).

TELEMETRY COSTS OF TRACES

One major consideration (and this topic could also fit in the instrumentation section previous, but I chose to talk about it here) is sampling of traces. We will discuss this topic in greater detail in chapter 8.

Sampling, in the context of distributed tracing, means to either capture or discard a certain trace a source emits. We usually differentiate between

- *Head-based (sometimes called up-front) sampling*—The agent discards a fixed number of traces. It's deterministic and helps when you're focusing on network costs; however, you may miss interesting traces (anomalies, errors, etc.).
- *Tail-based sampling*—The idea is to defer the sampling decision to when the trace is complete. But how do you know when the trace is done (stragglers)? It usually requires the agent to keep more state around and can lead to a more complex setup as well as higher resource usage (network + storage costs) compared to the simpler up-front sampling decision.

In addition to the runtime overhead (resource consumption) we have seen with logs and metrics already, there's also a tiny latency overhead in the communication path in the case of traces. This is due to the fact that context propagation costs time and bandwidth in the sources as well as on the wire.

DESTINATION COSTS OF TRACES

The distributed tracing solutions in general use data stores that allow you to deal with sparse, timestamped data. In general, traces need to be indexed so that you can search or filter for a particular service or can order by duration. This typically means the storage (backend) needs to be able to index certain span characteristics, such as service name, tags, and other metadata (e.g., the HTTP method).

There are different approaches used in the wild to realize trace storage and retrieval, which can be grouped as follows:

- Dedicated data stores, such as ClickHouse (https://clickhouse.com/)
- General-purpose NoSQL data stores, such as Cassandra, HBase, or Elasticsearch/OpenSearch, that are used with domain-specific schemas and logic to accommodate needs
- Object storage–based backends (e.g., using S3 buckets) that come with querier/ingester logic, such as with Grafana Tempo

The costs of the backends in tracing, like metrics, usually fall into one or more of three buckets: ingestion (how many traces with what payload are inserted), processing (sometimes called *scanning* or *traces retrieved*), and long-term storage costs (retention policies). We will discuss backends in detail in chapter 5.